What changed after the submission

A running record of every change to NeuroMPS after the round-1 deadline (2026-10-10 15:59:59 UTC; the submitted state is commit `464eb76`). Each entry opens with the one sentence a slide would carry, then says what changed, why, what it did to the results, and where the evidence is. Numbers cite `outputs/metrics.json` as the report does, so the citation check holds this document to the same standard. Newest entries are at the end; planned work is in the queue at the bottom.

#	Date	Change	Effect on results	Commits
1	2026-10-10	Unused Kaggle notebook removed; seven figures redrawn at print size	none (no number or claim changed)	`e5ed404`, `bbd75f9`
2	2026-10-11	Bicuculline doses put on one unit scale	the dose conclusion reverses	`886d8d3`
3	2026-10-22	Demo app runs as a hosted page from the repository alone	none; access only	`c4a9004`
4	2026-10-22/23	From chip to animal: the drug knobs on a whole fly brain	new results, pre-registered	`c83eb14` … `aba184c`
5	2026-10-24	Report page limit lifted; the fly study joins the report	none; structure only	`1c7d94f`
6	2026-10-24	EPA screen cutoffs set beside the fly's precision needs	new comparison (descriptive)	`cd074b3`
7	2026-10-24	Vesicle depletion tested as the cause of the GABA-A reversal	registered prediction failed; new negative result	`3071ab4`, `8fb5065`
8	2026-10-24	The twin as a source of labelled data, and as a prior for the classifier	1 of 3 registered predictions pass	`761068e`, `93bb4f6`
9	2026-10-24	Electrodes on the fly brain's output tract against a planar array	4 of 4 registered predictions pass	`cd074b3`, `2c66067`
10	2026-10-24	Behaviour read at the motor neurons of the male CNS	6 of 7 registered predictions pass	`a1c5f27`, `a2f4ee2`
11	2026-10-24	The motor-level ties resolved on a finer dose grid	1 of 2 registered predictions pass; one tie resolves	`69db4e6`, `796a33b`
12	2026-10-24	Roadmap to an embodied, closed-loop fly	plan, no new results	`42ee520`
13	2026-10-24	Embodied stage 1: the male CNS drives a muscle-driven leg	both registered tests fail; finding the cause changes the plan	`cbdd6d6` … this branch
14	2026-10-24	Embodied stage 1c: the male nerve cord from descending neurons, brain removed	3 of 5 registered predictions pass	`739f6f0` … `e576cd9`
15	2026-10-24	Embodied stage 1d: a held reach against a rhythmic swing, fresh seeds	4 of 4 registered predictions pass; D2 (the swing) fails under the audit's corrected measure (entry 22)	`66502eb` … `19ab0ae`
16	2026-10-24	Embodied stages 1e and 1f: the FlyWire brain drives the male nerve cord and the leg	1e uninterpretable as registered; 1f 5 of 5 pass on fresh seeds	`f2f5ad3` … this branch
17	2026-10-24	Embodied stage 2a: fictive walking in the male nerve cord	1 of 5 pass; no walking rhythm	`91c5b9e` … `ff43970`
18	2026-10-24	Embodied stage 2b: sugar extends the flybody proboscis, bitter holds it back	6 of 6 registered predictions pass	`76f90e8` … `4b586c0`
19	2026-10-24	Embodied stage 2c: a registered adaptation sweep	stopped by its rule; no cell walks	`c4c29a1` … `5610ae8`
20	2026-10-24	Embodied stage 2d: the glutamate sign rule	1 of 7 at the selected cell	`edbf611` … `a0ccff7`
21	2026-10-24	Embodied stage 2e: the Pugliese et al. rate model; a flaw in our rhythm measure	2 of 8 (one vacuous); the size stand-in starves DNg100	`21df803` … `7a13eba`
22	2026-10-24	The registered rhythm-measure audit	one earlier pass (1d D2) withdrawn; all else stands; originals reproduce	`76611c0` … this branch

#	Date	Change	Effect on results	Commits
23	2026-10-24	Embodied stage 2f: command neurons driven alike, two size rules	precondition fails in both; the command is silenced by the activity it starts	`7e36cfb` … this branch
24	2026-10-24	Embodied stage 2g: the published front-leg network, rebuilt from the male CNS	precondition fails narrowly (12 and 15 of 32); exploratory: a regular motor rhythm in the feasible draws	`00cefa1` … this branch
25	2026-10-24	Embodied stage 2h: the front-leg network with a fine input search	4 of 7 (synapse-count sizes) and 5 of 7 (no size scaling) pass, one of them vacuously in each; a leg rhythm under DNg100; circuit-dependent without size scaling; no alternation	`c6d7874` … this branch
26	2026-10-24	Embodied stage 3a: the first closed loop (claw-neuron feedback, grooming leg)	3 of 5; go/no-go fails: feedback shifts motor pools toward resistance but barely moves a held leg, and a sham does the same	`5cdeba9` … this branch
27	2026-10-24	Embodied stage 3b: a push on the tibia during the grooming hold	2 of 5; the loop does not speed the return; the front leg's proprioceptors are thinly traced	`cd0a711` … this branch
28	2026-10-24	Embodied stage 4a: the drug design with movement as the readout	3 of 6; the chip flags after movement fails in all five families; the tract no longer leads	`b1a49df` … this branch
29	2026-10-24	Embodied stage 5: simulated directions against published fly findings	no qualifying pharmacology found (both predictions uninterpretable); identified-neuron studies agree in 1 of 3	`1723e0c` … this branch
30	2026-10-24/25	Release: one command reproduces every fly number; report Act VIII, paper draft, decision log	reproduction identical; precision at movement onset (exploratory); the stage-2d corrected cell fails on fresh seeds (2 of 7)	`f053de3` … this branch
31	2026-10-25	Embodied stage 5b: the identified 13A/13B leg neurons, manipulated	2 of 5; silencing 13A shortens the reach as published; 13B and activation do not	`c20fa08` … this branch
32	2026-10-25	Embodied stage 5c: the 13A/13B manipulations read at the joint	0 of 4; the model's inhibitory leg neurons brake rather than steer; stage 5b's agreement came by another route	`04a2387` … this branch
33	2026-10-25	Embodied stage 3c: the perturbation test on the well-traced hind leg	2 of 4; strong feedback destabilizes the hold; resistance beats the sham (exploratory)	`22b082e` … this branch
34	2026-10-25	Embodied stage 3d: the hind-leg contingency test, paired, fresh seeds	0 of 2; no push-following effect; stage 3 stays a documented negative	`8ea194a` … this branch

#	Date	Change	Effect on results	Commits
35	2026-10-25	Embodied stage 4b: the movement-level headline on fresh seeds	3 of 4; the chip is late in every family again; without the bitter-gated proboscis one family ties	`b23f6a3` … this branch
36	2026-10-25	Embodied stage 4c: a finer dose grid inside the one-step margins	3 of 4; excitability block resolves chip later; GABA-A potentiation's lateness rests on the bitter-gated proboscis	`f387845` … this branch
37	2026-10-25	Embodied stage 2i: the front-leg rhythm with the published size rule, neuron volume	6 of 7; the rhythm needs the three-neuron circuit; promotor and remotor keep a reliable offset, not antiphase; flexors silent	`d03ee27` … this branch
38	2026-10-25	Embodied stage 2j: shuffled volumes and a whole-cord normalization	2 of 4; shuffled volumes give no rhythm; the normalization changes nothing; the promotor-remotor offset does not replicate	`1d1d644` … this branch
39	2026-10-25	Embodied stage 4d: the walking rhythm as a drug readout	2 of 4; acetylcholine block is the first chip-first case; GABA-A potentiation spares walking; the four-movement index stays chip-late	`73ffde6` … this branch
40	2026-10-25	Embodied stage 4e: the walking readout as a fraction of rhythmic draws	2 of 4; GABA-A and GluCl block break walking first again; the acetylcholine exception depends on the readout and is withdrawn	`180476f` … this branch
41	2026-10-25	One command reproduces every fly number again; the paper draft moves to a private repository	every recorded value reproduced; documents only	this branch
42	2026-10-26	Real ToxCast vehicle wells: noise floors, wells needed, and the twin against unseen positive controls	new real-data results, both registered; twin 12 of 12 directions, no better than a firing-rate rule	`cf90872` … this branch
43	2026-10-27	The registration audit and hash chain, in this repository	every registered prediction counted; no result file precedes its registration; corrected 2026-10-28 to count two studies it had missed	`d183585` … this branch
44	2026-10-28	The twin's knob dose-response against real concentration-response curves	2 of 4 registered predictions pass; the mechanism label matters, the curve shape does not clear the line	`f3e1fb9` … this branch
45	2026-10-27/28	The twin's inverse model recalibrated: 5.8x the simulations and a five-flow ensemble (D31)	2 of 7 registered predictions pass; on fresh simulations the round-1 model fails all five parameters; the ensemble passes only `threshold_scale` under the registered rule	`0bc29aa` … this branch

#	Date	Change	Effect on results	Commits
46	2026-10-28	Human iPSC against rat chips from one lab: split-half noise of untreated wells	2 of 3 registered predictions pass; human firing rate about twice as noisy, but not across readouts; wells needed reported as a lower bound	`35df26a` … this branch
47	2026-10-28	Our extractor rerun under the Kapucu 2022 depositors' stated settings, one change at a time (post hoc); the feature definitions recorded as D33	none to earlier results; the twofold burst-rate difference is traced mainly to the burst-rate denominator, on the hPSC plates; two report wording errors fixed	`46d1607`, `e10a2ff` … this branch
48	2026-10-28	The depositors' logISI burst detector, implemented and tested against their meaRtools output (D34, registered)	1 of 4 registered predictions pass; burst duration and spikes per burst reach their scale but not their ordering, the hPSC burst rate overshoots; the detector attribution stays open	`e11bc05`, `9570a9f`, `6f63385` … this branch
49	2026-10-28	Settling or steady noise in the human iPSC split-half spread (D35)	3 of 3 registered predictions pass; the registered linearity check fails for human wells; the human spread is mostly uneven settling, and five more minutes would not remove it	`e743c4c`, `bc6a56b` … this branch
51	2026-10-28	Named definition sets: the asset's spike-time features under ours and the Kapucu 2022 depositors' stated definitions, in a sidecar table, with a translation table (D37)	none to earlier results (asset and metrics byte-identical); three features carry over as the same number, three by a plate-dependent factor, burst rate not at all	this branch

1. Housekeeping pushed just after the deadline

Slide message: none; listed for completeness.

Two commits reached GitHub after the deadline: one removed a Kaggle notebook nothing used (`e5ed404`), the other redrew seven figures at the size they print at (`bbd75f9`). They were reverted during round-1 judging so that reviewers saw the submitted state (`2ecca42`), and restored after it (`a05b013`).

2. A unit error hid a dose trend, and another team's result exposed it

Slide message: *Burst rate falls with bicuculline dose after all; we had ranked nanomolar doses as if they were micromolar, and found the bug by taking a competitor's disagreement seriously.*

- **What changed.** ScreenNeuroPharm records its two lowest bicuculline doses in nM and the rest in µM. Three analyses ranked the dose value without its unit, so 0.1 µM sat among the highest doses. One function, `models/dataset.doses_in_um`, now converts in all of them, and a test (`tests/test_dose_units.py`) checks each.
- **Why.** NeuroChip Twin, another entry, reported a bicuculline dose trend that our report said did not exist. Tracing the disagreement found the bug.

- **Effect.** The feature dose test moved from no feature tracking dose to 2 of 7: burst rate (Spearman −0.5813) and network-burst rate (−0.604) fall with dose across 113 recordings. The twin's inferred inhibition still does not follow dose, so the conclusion became: the measurement carries the trend, and the inverse model is what fails.
- **Evidence.** Report §18 (negative results) and Appendix M; `src/neuromps/models/dataset.py`.

3. The demo runs as a hosted web page

Slide message: *Judges can open the demo in a browser; nothing to install.*

- **What changed.** Streamlit Community Cloud installs a requirements file and runs `app/app.py` without installing the package. The app now finds `neuromps` in `src/`, and `app/requirements.txt` pins only what the page needs, at the project's versions (a test keeps them equal). The one live-simulation button says it needs the full local install instead of failing.
- **Effect.** None on results. Checked in a clean environment with only the hosted requirements: every tab renders.
- **Evidence.** `tests/test_app_hosted.py`.

4. From chip to animal: the chip's drug knobs on a whole fly brain

Slide message: *On a simulated whole fly brain, the chip's firing rate ranks drugs correctly but raises the alarm late: in every drug family, behaviour failed first.*

- **What changed.** A new package, `src/neuromps/fly/`, runs the connectome-constrained model of Shiu et al. (2024) on FlyWire with 138,639 neurons, applies the twin's drug knobs by transmitter class, and reads the brain both as a chip (a virtual electrode array scored by `features/mea.py`) and as an animal (the neurons the published model validated against feeding, aversive gating and grooming). It is replicated on the male CNS connectome (165,122 neurons). New data sources, all CC-BY or MIT, enter the manifest as optional; `run_all.py --stage fly` runs it, and the app has a fourth tab.
- **Why.** Round 1 scored us lowest on practical value. A chip assay is used to predict what a drug does to a nervous system, which no culture can check; a mapped, simulated brain can.
- **How it was kept honest.** Three pre-registrations, each committed before the simulations it scores; the release of every table checked against the model's (the published neuron lists come from an older release); every failed prediction reported.
- **Effect.**
- The model is the published one: sugar drives the feeding motor neuron at 76.75 Hz against 77.375 Hz on the release the paper used.
- The chip's firing-rate change tracks behavioural disruption (Spearman 0.863; 0.878 on the male CNS).
- Registered prediction failed: the chip, at the registered threshold, flags the drug after behaviour has failed in all five drug families.
- At behavioural onset, GABA block has moved firing rate by 0.149 log2, about the untreated variation of the best real chips in the asset (0.156).
- A candidate early-warning feature (network-burst rate) proved to be noise: untreated recordings already cross its threshold in a fraction 0.382 of splits.
- Spontaneous activity recovers the direction of wiring between brain regions at chance (0.4946).

- **Evidence.** Report Act VII (§23–§26); `preregistration/`; figures `outputs/figures/fly_*_SIMULATED.png`; app tab 4.

5. The report has no page limit; the fly study joins it

Slide message: none; structure only.

- **What changed.** The 15-to-20-page gate on the report's main matter is removed from the build and from CI, now that the round-1 page limit no longer applies; a lower bound remains so a broken build that drops sections is still caught. The fly study moves from a companion note into the report as Act VII (§23–§26), after §22 so that no earlier section number changes; the short Appendix P that pointed to the note is gone.

6. A regulatory screen's hit cutoff, beside what the fly says a chip must see

Slide message: *EPA's MEA screen calls a firing-rate hit only beyond a change about six times larger than the one at which the fly's behaviour had already failed.*

- **What changed.** `src/neuromps/fly/screen_cutoffs.py` reads the hit cutoff of the ToxCast MEA firing-rate endpoints from the asset's 20,375 ToxCast records and converts it to log2. The acute screen normalises activity to a positive control at −100 (activity abolished); its cutoff of 70.97 means suppression to about 29% of control, a change of −1.784 log2, over 515 screened chemicals.
- **Why.** The review pointed out that most of the asset's records sat outside the modelling. This puts them to a use the fly study makes possible: does the screen's own threshold leave room to catch the mechanisms in time?
- **Effect.** The largest firing-rate change at which the fly's behaviour had already failed is 0.304 log2 (acetylcholine block), so the screen's cutoff is 5.87 times larger. Caveat: the cutoff applies to a fitted curve's maximum response across the tested range, not to one dose, so this compares magnitudes, not decisions. The fly side is simulated; the cutoffs are real.
- **Considered and dropped.** A compound-by-compound potency bridge between our spike-level sources and ToxCast: only four compounds appear in both (bicuculline, deltamethrin, acetaminophen, glyphosate), too few to rank.

7. Vesicle depletion does not explain why real chips slow down under GABA-A block

Slide message: *We tested the reviewer's best explanation for our twin's one wrong direction, registered in advance, and it does not hold: depletion shrinks the firing-rate rise but never turns it into the fall real chips show.*

- **What changed.** `src/neuromps/twin/depression_drugs.py` runs the four drug knobs and a five-step GABA-A dose series under each depression setting (Tsodyks–Markram depletion at the recurrent excitatory synapses), and scores a rule registered before the runs (`preregistration/twin_depression_under_block.json`).
- **Why.** The post-round-1 review proposed depletion as the missing mechanism behind three weak points at once: the twin's wrong firing-rate direction under GABA-A block (§14), its missed bicuculline dose trend (§18) and the cross-platform reversal (§11).
- **Effect.** Under strong GABA-A block the twin's firing rate rises by 2.389 log2 without depletion and by at least 0.8687 with it; real GABA-A antagonist pairs change by −0.335. None of the 7 settings qualifies, and burst rates do not fall with dose under any. Registered outcome: V1 fail, V2 pass, V3 not applicable. The weights were not recalibrated with depletion in place, so the mechanism is not ruled out, only depletion added to the present twin.
- **Evidence.** Report §18, row 9 and figure; `outputs/twin/depression_under_block_simulated.png`.

8. Trained on the twin alone, the classifier recognises real mechanisms

Slide message: *A classifier that never saw a real label recognises real drug mechanisms at two and a half times chance; the same simulations do not improve a classifier trained on real labels.*

- **What changed.** `src/neuromps/models/twin_prior.py` simulates 30 labelled pairs for each of the five mechanism classes the twin has a knob for, and tests two uses of them, registered before simulating (`preregistration/twin_prior_classifier.json`): as the centre of the classifier's penalty (the review's Fix 2), and as the only training data.
- **Effect.**
- Trained on simulated pairs alone and applied to the 73 real pairs: balanced accuracy 0.41, interval 0.2808 to 0.5072, against one in six by chance (pass).
- As a prior: balanced accuracy 0.53 against the baseline's 0.537, and conformal sets unchanged (two fails).
- A finding for free: real sodium-channel blockers are never recognised, because the twin's threshold knob moves firing by −3.273 log2 where the real blockers move it by −0.795: right direction, wrong size.
- **Evidence.** Report §14, last paragraph.

9. Record the tract, not the tissue around it

Slide message: *On the simulated fly brain, electrodes on the output tract (the brain's own microchannels) catch GABA block before behaviour fails; a planar array of the same size does not.*

- **What changed.** `src/neuromps/fly/tract.py` reads the parent sweep's spontaneous recordings through three arrays: the full planar array, 64 of its electrodes, and 64 single-unit electrodes on descending neurons. The test, at one threshold set by real-chip noise, was registered before recording (`preregistration/fly_output_tract_electrode.json`).
- **Why.** The review's Investigation B: does instrumenting the connection between compartments reveal failure a flat array misses? Our real devices (§15) had already said that only electrodes inside microchannels see direction.
- **Effect.** All four registered predictions pass. Under GABA block, at behavioural onset, the tract's firing rate has moved by 0.361 log2 against 0.044 for the equal-count planar array; the tract flags 3 families in time against 2, with no false alarms. Checks added after the pass: over ten random draws of each array the tract is ahead in 7 of 10, and the descending-neuron behaviour readout is not what drives it.
- **Evidence.** Report §24 ("Where to put the electrode") and figure; §20, fifth area.

10. One step closer to behaviour: the motor neurons

Slide message: *Read at the motor neurons that move the fly's mouthparts and legs, the chip is never ahead of behaviour: it is late in three drug families and level in two.*

- **What changed.** `src/neuromps/fly/motor.py` repeats the male-CNS replication's behaviour trials (same seeds) and reads them at the proboscis motor pool (feeding and its bitter gate), the front-leg motor pool (antennal grooming) and all motor neurons (recruitment). Registered before any drug condition was run (`preregistration/fly_motor_readout.json`).
- **Why.** The behaviour readouts were firing rates of interneurons; the male connectome includes the nerve cord, so behaviour can be read where it leaves the nervous system. This is the first step toward a closed-loop, embodied fly, which needs a body and sensory feedback on top.

- **Effect.** Six of seven predictions pass. The interneuron readouts reproduce exactly; sugar drives the proboscis pool at 31.405 Hz and bitter cuts it to 5.625; the chip tracks motor disruption (Spearman 0.8922). The chip-is-late prediction failed narrowly: late in 3, tied in 2, earlier in 0 (four late were needed). Motor output fails no later than the interneurons in every family.
- **Evidence.** Report §25 ("One step closer to behaviour") and figure; `outputs/fly/malecns_motor_SIMULATED.csv`.

11. Resolving the two ties, without re-scoring the failed test

Slide message: *We did not move the goalposts on the failed test; we ran a new registered one. With the ties resolved, the chip is late in four drug families, level in one, and never ahead.*

- **What changed.** `src/neuromps/fly/tie_break.py` runs four finer dose levels inside each of the two tied steps of entry 10, with the same seeds and untreated references, registered before running (`preregistration/fly_motor_tie_break.json`).
- **Why.** Entry 10's chip-is-late test failed because two families tied, both onsets inside one coarse dose step. Changing the test after the fact would have undone the registration; a new, registered test with a finer grid answers the open question honestly.
- **Effect.** The re-run tied levels reproduce exactly. The raised threshold resolves to chip later (motor at 0.32, chip at 0.4); GABA potentiation stays tied. The follow-up predicted both would resolve, so it failed. Entry 10's test stays failed; combined, the chip is later in 4, level in 1, earlier in 0.
- **Evidence.** Report §25 and figure; `outputs/fly/malecns_motor_tie_break_SIMULATED.csv`.

12. A roadmap, not a rushed demo: the embodied, closed-loop fly

Slide message: *Next, the simulated fly gets a body and its sensory feedback, so that "does the chip flag a drug before behaviour fails?" can be asked of movement itself; six stages, each with a registered go/no-go test.*

- **What changed.** `report/ROADMAP_EMBODIED_FLY.md` (Word and PDF beside it) sets out the plan: brain and nerve cord, motor neurons to muscles, muscles to joints in a physics body (NeuroMechFly v2 on MuJoCo), and joints back to proprioceptive neurons. A small script, `src/neuromps/fly/roadmap_inventory.py`, counts what the male connectome already provides: of 381 leg motor neurons, 260 are typed by the muscle they drive, and 1,030 proprioceptive sensory neurons can carry the feedback.
- **Why.** Building it properly takes weeks; showing a half-working body would cost credibility. A staged plan with registered tests shows where the project goes and how it will be checked.
- **Evidence.** `report/ROADMAP_EMBODIED_FLY.md`.

13. The leg moves, but not specifically: what stage 1 found

Slide message: *Wired to a muscle-driven leg, the male fly brain moves it toward the head under antennal drive, but under sugar drive too; the cause is that the published model's parameters do not carry over to the second connectome, where activity avalanches.*

- **What changed.** MuJoCo (an optional dependency) runs the FlyMimic leg from FlyGym 2; `src/neuromps/fly/embodied_stage1.py` maps male-CNS motor neurons onto its muscles by name and drives it open loop. Two registered tests: stage 1 and stage 1b (weight rescaled by the connectomes' input-synapse ratio).
- **Effect.** Stage 1 fails on specificity (sugar brings the claw closer than antennal drive); stage 1b's feeding check fails (sugar drives the feeding motor neuron at 0.0 Hz), so its leg results are uninterpretable. The cause: a sugar drive activates 251 neurons in the published FlyWire model and 14,263 in the male CNS.

- **What it changes.** The next step drives the male nerve cord from its descending neurons alone, then joins the validated FlyWire brain to it (roadmap §7).
- **Evidence.** `report/ROADMAP_EMBODIED_FLY.md` §7; `outputs/fly/embodied_stage1*_SIMULATED..`

14. The grooming neurons move the right leg

Slide message: *Driven from the descending neurons that start antennal grooming, the male nerve cord moves the front leg, and only the front leg, to the head and holds it there; walking neurons swing it instead.*

- **What changed.** `src/neuromps/fly/embodied_stage1c.py` cuts the male CNS to its nerve cord (brain removed), drives named descending neurons at 150 Hz, and moves the FlyMimic leg as in stage 1. Registered before running.
- **Effect.** Three of five predictions pass: silence without drive; the left front-leg motor pool at 15.846 Hz under grooming drive against 2.72 and 1.697 for the other legs; no other descending neuron recruits the front leg as strongly. The kinematic prediction failed: forward-walking drive dips nearly as close to the head (1.154 mm against 1.1345) at the bottom of a rhythmic swing. A sustained reach and a swing need a different readout, to be registered as stage 1d.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1c_SIMULATED..`

15. The fly's grooming neurons hold its leg at its head

Slide message: *On fresh seeds and with readouts registered in advance: driving the fly's antennal-grooming neurons brings the simulated front leg to the head and holds it there for nearly the whole second; escape neurons extend it. (That walking neurons "swing" it did not survive the rhythm-measure audit; see entry 22.)*

- **What changed.** `src/neuromps/fly/kinematics.py` adds two readouts (time near the head; rhythmicity), `src/neuromps/fly/embodied_stage1d.py` reruns stage 1c's design on fresh seeds. The readouts were suggested by stage 1c's traces, so they were registered before this run (`preregistration/embodied_stage1d_reach_versus_swing.json`) and tested out of sample.
- **Effect.** All four predictions pass: time near the head 0.9665 under grooming drive against at most 0.368 under any other; forward walking more rhythmic than grooming; the same at every activation scale; the front-leg specificity of stage 1c replicates. Stage 1c's failed predictions stay failed. *Found later (entries 21, 22): the rhythmicity measure had no peak requirement; under the registered audit's corrected measure D2 fails (forward walking 0.0956); the hold at the head (D1) does not depend on it.*
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1d_SIMULATED..`

16. From an antennal touch in the brain to a leg at the head

Slide message: *Antennal input to the validated FlyWire brain, relayed through matched descending neurons into the male nerve cord, moves a muscle-driven leg to the head; sugar input does not, and removing four grooming neurons abolishes the movement.*

- **What changed.** `src/neuromps/fly/model.py` can replay recorded spike trains into chosen neurons; `src/neuromps/fly/relay.py` pairs descending neurons across the two connectomes by cell type and side and joins the FlyWire brain to the male nerve cord; stage 1e and stage 1f were each registered before running.
- **Effect.** Stage 1e's absolute precondition failed (grooming descending neurons at 8.531 Hz against a 10 Hz line), so its leg results are uninterpretable. Stage 1f, on fresh seeds, passes all

five predictions: claw to 0.5369 mm under antennal input against 0.9377 under sugar; back to 1.4793 mm when aDN1 and aDN2 are removed.

- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage1f_SIMULATED.`.

17. The nerve cord alone does not walk

Slide message: *Driven by its walking command, the connectome-only nerve cord fires the leg extensors tonically and leaves the flexors silent: no rhythm. Rhythm needs intrinsic currents the connectome does not record; adding them is the next registered step.*

- **What changed.** `src/neuromps/fly/fictive.py` runs stage 2a (registered before running): the six legs' tibia flexor and extensor pools under walking, grooming and backward-walking commands, phases measured between pools. `src/neuromps/fetch/flybody.py` fetches the flybody whole-body model (Apache-2.0) for the feeding and walking body, each file checked against its git hash.
- **Effect.** Silent without drive (W0 passes); under the walking command the left middle-leg extensors fire at 54.12 Hz and the left front flexors at 0.0 Hz, so there is no rhythm and no phase (W1 to W4 fail). The roadmap listed this risk and the response to it.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2a_SIMULATED.`.

18. Sugar extends the proboscis, bitter holds it back

Slide message: *Sugar input to the validated FlyWire brain drives the motor neurons of each proboscis muscle the literature names, and a simulated proboscis extends fully; bitter input holds it back. Six of six registered predictions pass.*

- **What changed.** `src/neuromps/fly/proboscis.py` drives the flybody proboscis (body fixed, 0.01 ms step, checked against a step half that size); `src/neuromps/fly/embodied_stage2b.py` turns the proboscis motor pools (MN9, MN1/MN2, MN4, MN3, named by FlyWire cell type after McKellar et al. 2020) into rostrum and haustellum commands. Registered before running.
- **Effect.** Full extension under sugar (0.3431 mm of 0.3428); none under bitter, antennal input or none; time-averaged extension from 0.3054 to 0.0488 mm when bitter is added. Sugar recruits only feeding motor neurons (exploratory).
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2b_SIMULATED.`.

19. A registered sweep that found nothing, and why

Slide message: *We swept the one cell property the roadmap named, reported every setting, and none made the nerve cord walk. The wiring explains it: the walking command's active neurons inhibit the flexors about twice as much as they excite them.*

- **What changed.** `src/neuromps/fly/model.py` gains spike-frequency adaptation (off by default; with it off the spikes are identical to the published model's). `connectome.rewire` builds a degree-preserving control. `src/neuromps/fly/embodied_stage2c.py` runs the registered sweep, selection rule and confirmation. `src/neuromps/fly/flexor_diagnostic.py` is the exploratory follow-up.
- **Effect.** 0 of 36 cells produce a rhythm; the stage stops as registered. Flexors fire at 151.341 Hz once inhibition is removed; from the command's active neurons they receive 274.73 inhibitory against 130.78 excitatory synapses each (exploratory).
- **Why it matters for the interview.** It shows the discipline holding when the result is null: the sweep was fixed before running, reported whole, and followed by a mechanism, not by re-tuning until it walked.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2c_`.

20. The glutamate sign is not the missing piece

Slide message: *Making glutamate excitatory switches the silent flexors on, but the cord then fires as one slow wave, not as alternating legs. Three registered attempts agree that this wiring, in this neuron model, has no walking rhythm generator. That negative result is reportable.*

- **What changed.** `connectome.make_excitatory`; `src/neuromps/fly/embodied_stage2d.py` builds two registered bounds on where glutamate excites (onto motor neurons; everywhere), each without adaptation and over stage 2c's grid, through stage 2c's code (a rerun of stage 2c is byte-identical).
- **Effect.** Onto motor neurons: flexors 1.76 Hz, no cell passes a walking test. Everywhere: flexors 40.4 Hz with 7,118 active neurons, rhythm in 5 cells, alternation in none; the selected cell passes only silence without drive on fresh seeds.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2d_`.

21. The walking flies online are trained, not wired; and our rhythm measure was flawed

Slide message: *The walking flies circulating online are driven by trained controllers, not by connectome wiring. The one wiring-based walking rhythm (Pugliese et al. 2025) uses a firing-rate model. Our reimplementation did not test it fairly: our stand-in for neuron size starves the walking command. Checking the result exposed a flaw in our own rhythm measure, now being audited.*

- **What changed.** `src/neuromps/fly/rate_model.py` reimplements Pugliese et al.'s rate model from the paper and its public configuration, batched over parameter draws. `src/neuromps/fly/embodied_stage2e.py` runs the registered stage 2e.
- **Effect.** Under DNg100 drive, 11 neurons become active; DNg100 itself reaches 6.54 Hz, because size (here synapse count) sets its threshold. Rhythm predictions fail; the lesion prediction passes vacuously. DNb08's "rhythm" (0.7367) is one burst and a plateau. The registered rhythmicity measure has no peak requirement; the registered audit (entry 22) re-examined every stage that used it.
- **Why it matters for the interview.** We publish the flaw ourselves, say which earlier claims it touches, and fix it with a registered audit, not a quiet re-score.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2e_`.

22. The rhythm-measure audit: one registered pass did not survive

Slide message: *We found a flaw in our own rhythm measure, registered an audit before re-scoring anything, and re-ran every affected stage. One earlier pass (the walking leg's "swing") does not survive. Every other registered outcome stands, and the original numbers reproduce exactly.*

- **What changed.** `kinematics.rhythm` requires a genuine autocorrelation peak; `fictive.phase` and the period measures take a method switch (the registered measure stays the default); `src/neuromps/fly/rhythm_audit.py` re-simulates stages 1d, 2a, 2c, 2d and 2e on their own seeds and measures every trial both ways. Registered in `preregistration/rhythm_measure_audit.json` before any corrected value was computed on stage data.
- **Effect.** The registered measure reproduces every recorded outcome. Under the corrected one, stage 1d's D2 fails (forward walking 0.0956, grooming 0.2374); stage 2d's rhythm cells drop from 5 to 1; DNb08's "rhythm" in stage 2e falls to 0.0. Every other registered outcome is unchanged.
- **Why it matters for the interview.** A worked example of the discipline under pressure: the flaw found by ourselves, the fix registered before use, the earlier claim withdrawn in public.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/rhythm_audit_SIMULATED.json`.

23. The walking command is switched off by the activity it starts

Slide message: *Driven fairly, the walking command neuron starts a wave of activity that grows until it silences the command itself; the whole nerve cord then runs on its own. The published model avoids this by simulating only the front-leg network and capping recruitment at 500 neurons. That is what we test next.*

- **What changed.** `rate_model.drive_input` gives each driven neuron the input that alone brings it to a target rate; `src/neuromps/fly/embodied_stage2f.py` runs the registered stage 2f (two size rules, corrected rhythm measure, a precondition that the command is on, a lesion test that counts only if a rhythm exists).
- **Effect.** The precondition fails in both size rules (DNg100 0.18 and 0.0 Hz), so the other predictions are uninterpretable. Exploratory trace: DNg100 at 42.67 Hz after 50 ms, silenced by 0.25 s under -4452.5 inhibitory input; 5,247 neurons stay active without it.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2f_`.

24. The published front-leg network, rebuilt from the male CNS

Slide message: *Rebuilt as published, the male fly's front-leg network does produce a regular motor rhythm, but too few runs found a working input for the registered test to count. The network is nearly all-or-none, so the published input rule overshoots it. A finer input search is next.*

- **What changed.** `src/neuromps/fly/embodied_stage2g.py` builds Pugliese et al.'s front-leg network from the male nerve cord (2,133 neurons) and applies their recruitment rule, with the corrected rhythm measure.
- **Effect.** Precondition failed narrowly (12 and 15 of 32 draws; 16 needed), so the predictions are uninterpretable. Exploratory: in feasible draws, a regular low-rate oscillation in one or two motor-neuron pools (score 0.8747, period 154.0 ms without size scaling), removed by silencing the three-neuron circuit in that size rule only.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2g_SIMULATED.`.

25. The published walking rhythm, reproduced on the male CNS, with its limits

Slide message: *In our own reimplementation on a different fly's connectome, the walking command drives a front-leg motor rhythm at a walking fly's tempo. Without size scaling it needs the published three-neuron circuit, and not just as a relay. But it still does not alternate antagonist muscles.*

- **What changed.** `src/neuromps/fly/embodied_stage2h.py`: stage 2g's network with a geometric bisection of the input; `--check` adds the exploratory silencing variants.
- **Effect.** A working input in 32 and 18 of 32 draws; a leg rhythm in both size rules (0.8886 without size scaling, period 196.5 ms), removed by silencing the circuit only without size scaling. Exploratory: keeping the relay and silencing the other two neurons also removes it (0.0483). No reliable promotor-remotor offset, no tibia alternation; the grooming comparison passes vacuously.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2h_`.

26. The first closed loop: brain-free nerve cord, leg and leg sensors

Slide message: *The simulated nerve cord now moves a leg and feels it move. Feedback from the leg's position sensors shifts the motor neurons toward resisting the movement, as insect reflexes do. In a held posture that barely moves the leg, and blind input does the same. A perturbation test is next.*

- **What changed.** `body.Session` steps the leg; `model.simulate` takes an `extend` hook; `src/neuromps/fly/closed_loop.py` couples nerve cord and leg every millisecond and

drives sensory neurons from a joint angle; `src/neuromps/fly/embodied_stage3a.py` runs the registered arms; `--check` the exploratory activation-scale check.

- **Effect.** Go/no-go failed: tibia shift 0.0054 rad (0.1 needed); 3 of 5 pass (silence, no resistance from the opposite assignment, the reach survives). Motor level: extensor 43.25 to 31.75 Hz under resistance, matched by the sham.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage3a_`.

27. A push on the leg: the loop does not push back

Slide message: *We pushed the simulated leg and timed its return. The position sensors' feedback did not bring it back faster: the muscles alone set the return. The front leg's sensors are thinly traced in this connectome, about one fifth of the hind leg's. That is the limit to state, and to fix with a better-traced leg.*

- **What changed.** `body.Session.step` accepts joint torques; `closed_loop.Loop` takes a perturbation window; `src/neuromps/fly/embodied_stage3b.py` runs the registered perturbation test; `--check` counts proprioceptors per leg.
- **Effect.** Push flexes the tibia 0.628 rad; return 32.0 ms with resistance feedback against 30.0 open loop (2 of 5 pass; contingency uninterpretable). Left front leg: 4 claw neurons; left hind leg: 22.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage3b_`.

28. Drugs in the loop: the chip is still late when the readout is movement

Slide message: *We reran the drug study with the simulated fly's own movements as the behaviour. In every drug family, feeding and grooming change at a lower dose than the chip first notices. The chip is late at every level we have looked at: neurons, motor neurons, and now movement.*

- **What changed.** The stage-2b and relay trials take a drug knob (stage 2b reruns byte-identical); `src/neuromps/fly/embodied_stage4a.py` runs the parent's 22 conditions on the proboscis and the grooming reach and compares each family's movement onset with the recorded chip and tract detections.
- **Effect.** 3 of 6 pass: the movements are there under vehicle, acetylcholine block stops them, and movement never fails before the neural readouts. The chip flags after movement fails in all five families (GABA-A block: movement 0.2, chip 0.6); the tract no longer leads (4 families each); the bitter gate's release is not monotonic.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage4a_`.

29. Checking against real flies: the experiments have not been done

Slide message: *We registered how we would search the literature before searching, so we could not pick the papers that agree. No published study gives flies these drugs and scores feeding or grooming movements. The model makes predictions that real experiments have yet to test.*

- **What changed.** `preregistration/embodied_stage5_real_fly_directions.json` (queries, inclusion, mapping, comparison rule); `src/neuromps/fly/real_fly_literature.json` (every candidate with its classification); `src/neuromps/fly/embodied_stage5.py` scores the comparison against the stage-4a medians.
- **Effect.** 0 pharmacological or receptor-genetic comparisons, so both predictions are uninterpretable. Identified-neuron studies: the simulation agrees in 1 of 3 (agreement for silencing GABAergic grooming neurons; mismatches explained by class-wide knobs and the absence of satiety).
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage5_SIMULATED.json`.

30. One command reproduces every fly number; the report, a paper draft and a decision log

Slide message: *Every fly result, from the first brain simulation to the closed loop, now reruns with one command and comes back identical. The technical report carries the embodied fly, a paper draft exists, and every choice of path is logged so other branches can be explored.*

- **What changed.** `src/neuromps/fly/pipeline.py` and the `embodied` stage of `run_all.py`; Act VIII of `report/TECHNICAL_REPORT.md`; `report/PAPER_DRAFT.md`; `report/DECISION_LOG.md`; the README's embodied-fly section.
- **Effect.** A from-scratch run took about three hours and reproduced every recorded value (rerun with the later studies: entry 41). Exploratory: at movement onset the chip has moved 0.149 to 0.452 log2. The stage-2d cell the corrected measure selects fails on fresh seeds (rhythm 0.2035).
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage4a_precision_SIMULATED.json`, `outputs/fly/embodied_stage2d_confirm_SIMULATED.json`.

31. Manipulating the same neurons as the experiment: one of four directions reproduced

Slide message: *Silencing the fly's GABAergic 13A leg neurons in the simulation shortens the grooming reach, as silencing them reduces grooming in real flies. The other three published manipulations do not show in the model's reach, which points to what the model still lacks.*

- **What changed.** `src/neuromps/fly/embodied_stage5b.py` silences (outputs removed) or activates (150 Hz) the left front leg's 13A or 13B neurons in the grooming relay, registered in `preregistration/embodied_stage5b_identified_neurons.json` before running.
- **Effect.** 2 of 5 pass: the reach is there; silencing 13A shortens it to 0.3806 mm from 0.9927; 13B leaves it unchanged; activating 13A shortens it only to 0.7309 mm.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage5b_SIMULATED.`.

32. At the joint, the model's leg neurons brake rather than steer

Slide message: *Read at the knee, none of the published leg postures appears when the simulated 13A/13B neurons are silenced or activated, and the one earlier agreement came by another route. The model's inhibitory leg neurons act as broad brakes, not joint-specific controllers: a stated limit of the connectome-only nerve cord.*

- **What changed.** `src/neuromps/fly/embodied_stage5c.py` reads stage 5b's manipulations at every front-leg joint, registered in `preregistration/embodied_stage5c_identified_neurons_joints.json` before running.
- **Effect.** 0 of 4: silencing 13A extends the tibia (-0.1488 rad) instead of flexing it; activating 13A relaxes it toward rest (+0.8325 rad) instead of over-extending it.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage5c_SIMULATED.`.

33. On a well-traced leg the sensors act, and destabilize the hold

Slide message: *Closed through the hind leg's 25 position sensors, feedback is strong enough to change the leg, but it makes the hold swing instead of steadying it. Against blind input the feedback that follows a push may speed the return; a registered test of that alone is next.*

- **What changed.** `src/neuromps/fly/embodied_stage3c.py`: a 1 ms closed loop for flybody's servoed right hind tibia, its 25 claw neurons, and stage 3b's perturbation test; registered in `preregistration/embodied_stage3c_hind_leg_reflex.json`.

- **Effect.** 2 of 4: held and pushed as required; resistance feedback returns the leg in 23.0 ms against 19.0 open loop (fail); the extensor pool falls from 34.0 to 18.0 Hz; against the sham, 23.0 against 32.5 ms (exploratory).
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage3c_SIMULATED.*`.

34. The registered retest: no push-following effect on the hind leg

Slide message: *We registered the one hopeful sign from the hind leg as its own test, on fresh seeds and more of them, and it did not hold. The loop stays a documented negative, with the body model it would need named.*

- **What changed.** `src/neuromps/fly/embodied_stage3d.py`: stage 3c's closed loop on 32 fresh seeds, resistance against sham paired by seed; registered in `preregistration/embodied_stage3d_hind_leg_contingency.json`.
- **Effect.** 0 of 2: resistance returns sooner than the sham in 14 of 32 seeds (22 needed); median paired difference 0.0 ms.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage3d_SIMULATED.*`.

35. The headline, rerun: the chip is still late, by one dose step in two families

Slide message: *We reran the paper's main result on fresh seeds against the chip's own replication: movement still fails first in every family. Leaving each movement out shows where the margin is thin, and we report that rather than smooth it over.*

- **What changed.** `src/neuromps/fly/embodied_stage4b.py` reruns stage 4a's movements on 16 fresh seeds (stage 4a's trial loop now lives in reusable functions; its recorded trials rescore to the same outcomes); registered in `preregistration/embodied_stage4b_movement_replication.json`.
- **Effect.** 3 of 4. The chip flags later than movement fails in all five families (GABA-A block 0.6 against 0.2), and every movement onset is within one dose step of stage 4a's. The leave-one-out check fails: without the bitter-gated proboscis, GABA-A potentiation impairs movement only at 1.0, where the chip flags it.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage4b_SIMULATED.*`.

36. Inside the thin margins: one resolves, one rests on a single movement

Slide message: *Where the coarse dose grid left the chip late by one step, a finer grid keeps it late. In one family that lateness comes from the proboscis movement that bitter gates, and we say so.*

- **What changed.** `src/neuromps/fly/embodied_stage4c.py`: GABA-A potentiation and excitability block at finer levels, movements on stage 4b's seeds, the chip on both of the parent's seed sets, the coarse levels rerun as a check (reproduced to floating-point rounding); registered in `preregistration/embodied_stage4c_fine_dose_grid.json`.
- **Effect.** 3 of 4. Excitability block: movement fails at 0.24, the chip flags at 0.36 or 0.4. GABA-A potentiation: movement at 0.5, chip at 0.6; without the bitter-gated proboscis both at 0.6 (F2 fails).
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage4c_SIMULATED.*`.

37. With the published size rule, the walking circuit is necessary

Slide message: *Using the neuron volumes the published model calls for, the walking command's front-leg rhythm needs the published three-neuron circuit on a second connectome. Our earlier stand-in for size had hidden that. The leg's antagonist muscles still do not alternate.*

- **What changed.** `src/neuromps/fetch/malecns_sizes.py` fetches the male CNS v1.0 neuPrint neuron table (4.65 GB, CC-BY), checked against the bucket's CRC32C by

`src/neuromps/fetch/crc32c.py` (numpy, no new dependency); `fly/malecns.neuron_sizes` reads its size column. `src/neuromps/fly/embodied_stage2i.py` reruns stage 2h with size = volume over the network median, fresh draws; registered in `preregistration/embodied_stage2i_volume_size_rule.json`.

- **Effect.** 6 of 7: rhythm 0.9446, gone with the circuit silenced (0.0115); promotor and remotor phase-locked (R 0.843) with remotor slightly ahead, not in antiphase; flexor-extensor alternation fails.
- **Evidence.** Roadmap §7 and figure; `outputs/fly/embodied_stage2i_SIMULATED.`.

38. The identified neurons' volumes matter; one hopeful detail does not hold

Slide message: *Shuffle the same volumes across neurons and the walking rhythm disappears, so it is the identified neurons' sizes that matter. The fixed coxa offset we saw once did not replicate, and we do not claim it.*

- **What changed.** `src/neuromps/fly/embodied_stage2j.py` (fresh draws, three size arms); the rate model takes a size per neuron and draw; registered in `preregistration/embodied_stage2j_volume_specificity.json`.
- **Effect.** 2 of 4. Volume rule: rhythm 0.6992 and circuit necessity replicate, the offset does not (R 0.6082). Shuffled: no rhythm (0.0021). Whole-cord median: unchanged (rhythm 0.9531).
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage2j_SIMULATED.`.

39. Read through walking, one drug is caught in time

Slide message: *With a walking rhythm as the behaviour, the chip still flags most drugs late, but acetylcholine block is caught before walking fails: which behaviour a chip stands in for decides which drugs it catches in time.*

- **What changed.** `src/neuromps/fly/embodied_stage4d.py`: the drug knobs on stage 2i's walking network (`rate_model.weights` takes per-class scales); registered in `preregistration/embodied_stage4d_walking_readout.json`.
- **Effect.** 2 of 4. GABA-A block breaks walking at 0.2; acetylcholine block: chip 0.4, walking 0.6; GABA-A potentiation spares walking; with walking as a fourth movement the chip is late in every family.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage4d_SIMULATED.`.

40. The exception we put in the abstract did not survive its retest

Slide message: *We retested our one chip-first case with a readout that fits the data, and it went away: mild acetylcholine block makes the walking rhythm more reliable, and that counts as a change before the chip flags. We took it out of the abstract.*

- **What changed.** `src/neuromps/fly/embodied_stage4e.py` (stage 4d's run loop, now `embodied_stage4d.run_trials`, on fresh draws, walking scored by the fraction of rhythmic draws); registered in `preregistration/embodied_stage4e_walking_fraction.json`. The paper abstract and discussion, the technical report, the README and the slides no longer claim the exception.
- **Effect.** 2 of 4. GABA-A and GluCl block break walking before the chip (E2). Acetylcholine block changes walking at 0.2, before the chip's 0.4 (E1 fails); by stage 4d's median readout it is chip-first again on these draws.
- **Evidence.** Roadmap §7; `outputs/fly/embodied_stage4e_SIMULATED.`.

41. Every fly number reproduces again; the paper is kept private until submission

Slide message: *Rerun from scratch after seven more studies, one command reproduced every recorded number and every figure byte for byte.*

- **What changed.** `python run_all.py --full --stage embodied` rerun on the current code (48 pipeline steps). The paper draft, its outline and the slide outline were removed from this repository; they are kept, with any new research toward the paper, in a private repository until the paper is submitted (decision D28).
- **Effect.** The stage took 15052.3 s. Every value in `outputs/metrics.json` reproduced except the run time itself; every output table, figure and JSON file was byte-identical; the data manifest changed only in its re-verification timestamps.
- **Evidence.** Roadmap §7 (stage 6); `outputs/metrics.json`.

42. How many wells a chip needs, from 714 real untreated wells

Slide message: *In the EPA's own MEA screen, untreated firing rate falls by a third and the day explains two-fifths of the noise: judged against same-day vehicle wells, 4 wells see a halving; judged against a historical reference, no number of wells from one day does.*

- **What changed.** The well-level rows of the EPA's acute MEA screen are now read from the invitrodb MySQL dump (CC0; `neuromps.fetch.toxcast_mysql`, `neuromps.etl.toxcast_wells`), with the seven electrophysiology readouts the public dump lacks recovered and checked (`neuromps.etl.toxcast_recover`). Two analyses, each registered before any response statistic was computed: per-readout noise floors and wells needed (`src/neuromps/models/toxcast_noise_floors.py`, `preregistration/toxcast_noise_floors.json`), and the twin's knobs against the screen's positive controls (`src/neuromps/models/twin_toxcast_check.py`, `preregistration/twin_toxcast_positive_controls.json`; the twin's predictions were committed before the real side was computed).
- **Why.** Round-1 judges scored practical value lowest. A lab using a chip first needs to know how large a change it can see and how to lay out its plates; the screen's vehicle wells answer that directly, and its positive controls are a laboratory the twin has never seen.
- **Effect.**
- Over 714 vehicle wells on 97 dates, untreated firing rate falls by a median of −32% and the day carries 0.42 of its variance.
- For a fall of a fifth in firing rate: 32 wells across days, or 32 treated and as many vehicle wells on one day; from one day against a historical reference, more than 96 (even a halving is out of reach). For a halving: 3 across days or 4 on one day with same-day vehicles. A table gives every readout; burst shape and synchrony are several times quieter.
- The twin agrees with the positive controls on 12 of 12 scored directions (all four registered predictions pass), including firing rate under GABA-A block, which rises in these wells (bicuculline 1.124 log2) though it falls in our asset's other GABA-A antagonist pairs. A post-hoc rule that follows firing rate also scores 11 of 11, so the test shows the twin is not wrong at strong doses, not that it adds information.
- **Evidence.** Report Act IX (§32-§34), the table in §32, §20 item 6, §18 row 9; `outputs/figures/toxcast_noise_floors_real.png`, `outputs/figures/twin_toxcast_positive_controls.png`, `outputs/tables/toxcast_wells_needed.md`.

43. The registrations audited and hash-chained in this repository

Slide message: *Every registered prediction is counted by a script, and every result file is checked against the commit that registered it, here, in the public repository.*

- **What changed.** Two report tools come over from the manuscript repository, where they were first written: `neuromps.report.audit` counts the outcome of every registered prediction in `outputs/metrics.json` (now run at the end of every `run_all`), and `neuromps.report.timestamps` writes `preregistration/TIMESTAMPS.json` (each registration's checksum and first commit, a hash chain over them, and whether each result file came after its registration) and its counts to `metrics.json`. Report §21 describes both.
- **Why.** The final-round deck quoted registration counts and a hash chain that existed only in the private manuscript repository, so a reader of this repository could not check them.
- **Effect.**
- 46 registered studies here made 233 predictions: 105 passed, 88 failed, 39 were uninterpretable.
- 41 result files name their registration, and 0 of them came before it; 4 registrations are not covered because their results do not name them.
- No earlier result changes.
- **Correction (2026-10-28).** As first merged, the audit counted 40 studies and 208 predictions. It counted only predictions that carry an `outcome` word, and two studies record a pass flag instead: the twin against the ToxCast positive controls (entry 42) and the dose response (entry 44). Both now record an outcome as well (pass, fail, or uninterpretable when a registered minimum is not met; no verdict changes), and the audit now names any study it cannot read (`registration_audit.unrecognised`, empty) and leaves out, by name, the rhythm audit, which re-reads predictions already counted under their own stages (`registration_audit.excluded`). The counts above include both studies, and the inverse-model recalibration (entry 45), whose predictions sit one level down under `by_prediction`, a layout the audit now also reads.

44. The dose response: what the twin's knobs get right, and where the test broke

Slide message: *Asked to reproduce the shape of real concentration-response curves, the twin beats a wrong-mechanism knob by 0.47 in rank correlation and names the first-moving readout for 8 of 11 compounds, but fewer than half its curves clear the registered line — and the failure sits entirely at the top concentration, where this screen cannot tell a mechanism from a general loss of activity.*

- **What changed.** A third registered analysis on the EPA acute MEA screen (`src/neuromps/models/twin_toxcast_dose_response.py`, `preregistration/twin_toxcast_dose_response.json`, decision D30, report §35). The twin's four drug knobs were simulated over a seven-level grid on 12 chips and the curves committed before any treated well was read; concentrations are matched to knob levels by rank; scores are a Spearman correlation per compound and readout, which readout moves first, and two nulls.
- **Why.** §34 recorded the gap in §33: the positive controls are single saturating concentrations, so nothing tested the twin's dose response — the next step the closing slide of the deck names.
- **Effect.**
- 2 of 4 registered predictions pass over 13 compounds and 26 scored pairs.
- Passing: the labelled knob beats a knob the compound is not labelled with by 0.47 in mean Spearman, and the twin names the first-moving readout for 8 of 11 compounds.
- Failing: only 10 of 26 pairs clear the registered correlation line, and the follow-firing-rate rule of §33 is again ahead (0.52 against 0.34).
- The registered exclusion rule, which drops concentrations from the first one the plates' viability readout flags, excluded nothing: over forty minutes that readout barely moves. Post

hoc, dropping each compound's highest concentration raises the twin to 0.65, while dropping the lowest gives 0.22 and a middle one 0.32. The registered verdict stands; the post-hoc result says where to look next.

- **Evidence.** Report §35; `outputs/figures/twin_toxcast_dose_response.png`, `outputs/twin/toxcast_dose_response_predictions_simulated.json`, `outputs/metrics.json` key `twin_toxcast_dose_response`.

45. The inverse model, recalibrated: sharper, still over-confident

Slide message: *On 1,000 fresh simulations, the round-1 inverse model is over-confident on all five parameters; five times the simulations and a five-flow ensemble keep its sharpness and move it toward calibration, but only the excitability parameter passes the registered checks.*

- **What changed.** A registered test (decision D31, `preregistration/twin_inference_recalibration.json`, committed before any new simulation) of two remedies for the two parameters the round-1 posterior failed: the same flow retrained on a larger pool, and an equal-weight ensemble of five flows. 12,750 fresh training simulations finished before the registered 05:00 cut (pool 15,750), plus 1,000 fresh held-out runs. `src/neuromps/twin/recalibration.py` (seeds, chunked simulation, coverage), `src/neuromps/twin/recalibration_run.py` (training, evaluation, scoring), a step in `run_all.py`'s twin stage. The round-1 result and code path are unchanged.
- **Why.** The inverse-model slide withholds two of five parameters; whether that is a limit of the simulation budget or of the estimator decides what to try next.
- **Effect.** 2 of 7 registered predictions pass.
- The round-1 failure replicates, and widens: on fresh simulations the round-1 flow fails all five parameters (`inter_in_degree` p = 0.0001). Post hoc checks rule out a changed simulator (6 of 6 of its runs re-simulate identically) and the evaluation code (on its own held-out set it reproduces the round-1 passes, while its 50% intervals cover only 0.3933 to 0.47).
- The ensemble keeps contraction (passes; `threshold_scale` 0.742) but fails `inter_in_degree` on the 300-run test (p = 0.0056), is over-confident for four parameters by coverage, fails four of five on all 1,000 runs, and costs 1.953 s per posterior against the registered 1 s.
- Budget alone fixes neither failing parameter (`gabaa_scale` p = 0.0025).
- The check near the real recordings fails on `ampa_scale` (p = 0.0121).
- Under the registered adoption rule the slide may quote `threshold_scale` from the ensemble, and nothing else.
- **Evidence.** Report §16 (the follow-up after the calibration table), §18 row 6, Appendix G; `outputs/twin/hd14b_recalibration_simulated.png`; `twin_hd14b_recalibration` in `outputs/metrics.json`.

46. A human iPSC chip's firing rate is about twice as noisy as a rat chip's, in one lab

Slide message: *In one lab's untreated recordings, a human iPSC well's firing rate moved about twice as much between the two halves of one recording as a rat well's (R = 2.06, 95% interval 1.39 to 3.56), so on these data a human chip needs more wells for the same firing-rate change; the difference is not general across readouts.*

- **What changed.** A registered real-data analysis on the Kapucu et al. 2022 untreated wells (`src/neuromps/models/hpsc_split_half.py`, `preregistration/hpsc_split_half.json`, decision D32, report §36). Each untreated recording is cut in half, both halves go through `features/mea.py`, and the spread of the change between halves is compared between species on the lab's matched 48-well baselines, then on its 12-well developmental plates.

- **Why.** The practical-value result (§32, entry 42) gives wells needed for rat cortical chips only; a judge will ask about human iPSC chips, and the deposit has too few untreated wells recorded twice to repeat §32's design.
- **Effect.**
- P1 passes: on the baselines R for firing rate is 2.06 (1.39 to 3.56) over 24 human and 38 rat wells.
- P2 fails: the human wells are noisier on 4 of 8 readouts, one short of the registered 5; the rat wells are the noisier on the share of spikes in bursts.
- P3 passes on its point estimate, as registered: R = 1.54 on the developmental plates, but the interval (0.75 to 2.95) includes 1.
- Reported, not tested, and a lower bound: at least 7 human against 2 rat wells per arm for a 20% firing-rate fall. Not comparable with entry 42's numbers, which include handling and day-to-day variation.
- The registered firing-rate-overlap check keeps the direction: R = 4.62 on the 7 human and 14 rat wells with overlapping firing rates.
- No earlier result changes. The registration counts in §21 and entry 43 rise by this study's three predictions.
- **Evidence.** Report §36; `outputs/figures/hpsc_split_half_real.png`, `outputs/hpsc_split_half_REAL.json` (with the per-well table), `outputs/metrics.json` key `hpsc_split_half`.

47. Why the extractor disagrees with meaRtools on burst rate, and what we chose

Slide message: *On the same spike files, our burst rate is less than half the depositors' published meaRtools value (ratio 0.4423); rerun with their stated settings, most of the gap is one choice, dividing by the electrodes that burst rather than by every active one (ratio 0.944), and it is a human iPSC effect. Post hoc.*

- **What changed.** `features/mea.py` gains two definition options, off by default (`active_strict`, `burst_rate_denominator`); at their defaults every feature value and every method record is byte-identical, which a golden-value test pins. A new module, `src/neuromps/models/extractor_ladder.py`, runs inside the existing extractor-agreement stage and reruns our extractor on the depositors' spike files with their stated choices switched on one at a time: active above 10 spikes/min (S1), their noisy electrodes dropped for each DIV (S2), burst rate per bursting electrode (S3), each change alone, and an assumption check. The definitions themselves are recorded retroactively as decision D33. Two wording errors in the report are fixed: the executive summary said we had run the depositors' meaRtools pipeline (we ran our extractor on their spike files and compared with the output they published), and the §3 figure caption described a different plot.
- **Why.** The judges' question was why a reader should trust our extractor. §3 showed where it disagrees with an independent implementation but not why; a twofold burst-rate gap with no explanation reads as a possible bug.
- **Effect.**
- The ladder's first step is the published comparison, unchanged (check: true); the agreement numbers in §3 do not move.
- Pooled burst rate, median ratio ours/theirs: 0.4423 as published, 0.6998 at S1, 0.6967 at S2 and 0.944 at S3; the denominator alone gives 0.9087. The ladder closes a share of 0.9293 of the pooled log-ratio gap.

- It is an hPSC effect: hPSC_MEA1 moves from 0.3417 to 0.8579, hPSC_MEA2 from 0.3841 to 0.8973, and the rat plate from 0.9573 to 1.04. In a median hPSC_MEA1 recording a share of only 0.6225 of active electrodes burst, against 0.9815 on the rat plate.
- With the same electrodes (S2), firing rate, active-electrode count and STTC reproduce theirs: rank agreement 1.0 for firing rate, and an identical active-electrode count in a share of 1.0 of recordings. Burst duration and spikes per burst barely move (S3 ratios 0.8208 and 0.9433): that remainder is the burst detector, which is not tested.
- The assumption that the noisy-electrode file is read per DIV column was checked: read as "listed on any DIV, noisy throughout", the active-electrode count matches in a share of only 0.253 of recordings.
- Post hoc and not registered: a scratch analysis showed these numbers before the module was written, so the registration counts (§21, entry 43) do not change. No earlier result, record or twin output changes; the asset is not rebuilt.
- **Evidence.** Report §3 ("Where the difference comes from"); decision D33; `outputs/figures/extractor_settings_ladder_real.png`; `outputs/metrics.json` key `extractor_agreement.settings_ladder`; tests `tests/test_features_mea.py`, `tests/test_extractor_ladder.py`.

48. The depositors' burst detector, tested: it does not explain the rest of the gap as registered

Slide message: *With the depositors' own adaptive burst detector (logISI) added to their stated settings, our burst duration and spikes per burst land on their scale (pooled ratios 0.96 and 1.00) but still rank recordings differently, and the human burst rate overshoots theirs; one of four registered predictions passes, so the detector does not account for the remaining gap, or our implementation differs from theirs.*

- **What changed.** `features/mea.py` gains a third definition option, off by default: `FeatureParams.burst_detector = "logisi"`, the adaptive detector of Pasquale et al. 2010 with the depositors' stated minimum of five spikes and merging of close bursts. At the default every feature value and method record is byte-identical (the golden-value tests pass). A new module, `src/neuromps/models/extractor_logisi.py`, runs inside the extractor-agreement stage: S4 is the ladder's S3 with logISI (defined in `extractor_ladder.py` beside the ladder, outside its definition set). The registration (`preregistration/extractor_logisi.json`, decision D34) fixed four predictions and the uninterpretable rule; its implementation section was completed and committed alone (`9570a9f`) before the detector ran on any Kapucu recording.
- **Why.** Entry 47 attributed what the settings ladder left (burst duration, spikes per burst, part of the hPSC burst rate) to the burst detector without a test; a measured attribution answers "why trust your extractor" better than an inferred one.
- **Effect.**
- L1 `fail`: pooled burst duration ratio 0.9611 is in its band, but rho 0.8347 is below 0.9 (S3: 0.8167).
- L2 `fail`: pooled spikes per burst ratio 0.9986 is in its band, but rho 0.9467 is below 0.95 (S3: 0.9443).
- L3 `uninterpretable`: the fallback covers a share of 0.515 of the hPSC active electrode-recordings, more than the registered half; the ratios, 1.0847 (hPSC_MEA1) and 1.1438 (hPSC_MEA2), lie above the band, where S3 lay below it.
- L4 `pass`: on the rat plate burst rate 0.9816 and burst duration 0.9825, up from 0.7705 at S3.

- The registered assumption check: fallback shares 0.4982 (hPSC_MEA1), 0.5321 (hPSC_MEA2) and 0.2313 (Rat_MEA1); median valid thresholds 7.08 ms, 8.91 ms and 281.84 ms. A share of 0.2786 and 0.2133 of the valid hPSC thresholds are below a millisecond, where a peak of near-coincident spikes is taken as the intraburst peak; this share was added to the output after the quick run showed such thresholds, and is descriptive.
- Post hoc, as the registration named it: from S3 to S4 the burst rate changes by more than a tenth in a share of 0.8273 (hPSC_MEA1) and 0.7632 (hPSC_MEA2) of recordings, the rat burst duration in 0.8723.
- The ladder's numbers and the published agreement are unchanged (every other key of `extractor_agreement` is identical to the previous run). The registration counts in §21 and entry 43 rise by this study's four predictions. The depositors' analysis scripts, shared on G-Node GIN, were not read; they are the next check if the attribution matters.
- **Evidence.** Report §3 ("The detector, tested"); decision D34; `outputs/figures/extractor_logisi_real.png`, `outputs/extractor_logisi_REAL.json` (with every electrode's threshold), `outputs/metrics.json` key `extractor_agreement.logisi`; tests `tests/test_features_mea.py`, `tests/test_extractor_logisi.py`.

49. The human iPSC split-half noise is mostly uneven settling, not steady noise

Slide message: *Within one untreated 30-minute recording, almost all of a human iPSC well's split-half firing-rate change is its own trend (share 0.996), and human wells differ far more than rat wells in that trend (SD ratio 3.65); so on these data a longer settling period before dosing is the first thing to try, but the recordings are too short to say how long it must be.*

- **What changed.** A registered follow-up to entry 46 on the same Kapucu et al. 2022 baselines and usable wells (`src/neuromps/models/hpsc_settling.py`, `preregistration/hpsc_settling.json`, decision D35, report §36, "Registered follow-up"). Each recording is cut into six 5-minute windows; per well, the firing rate on one fixed electrode set gives a straight-line slope (its settling) and a residual SD (its scatter), and the split-half change splits into a trend part and a residual part.
- **Why.** Entry 46 noted that part of the human spread might be wells settling at different speeds; for an assay the two readings call for different fixes (a longer settling period against more wells).
- **Effect.**
- T1 passes: the slope's SD is 0.076 human against 0.021 rat (log2 per window), ratio 95% interval 2.72 to 5.04.
- T2 passes: median residual SD ratio 10.2 (7.91 to 11.43), but most of it is curvature a straight line misses (next point), so it is not read as tenfold steady noise.
- T3 passes: the trend's share is 0.996 human and 0.946 rat. The registered threshold sits below what a line absorbs from pure steady noise (about 0.77), which we found when implementing it; the human share is well above that reference.
- The registered linearity check fails for human wells: a quadratic term lowers the residual SD by more than a third in a share of 1.0 of them (0.26 of rat wells), so T1 and T3 are read with that caveat. The human median rate falls from 6.73 to 4.26 Hz by the fourth window, then partly recovers.
- Post hoc: dropping the first window leaves the human spread 3.16 times the rat one, so five more minutes of settling would not close the gap. The residual-only wells needed (1 human well for a 20% fall) are illustrative only and are not used.

- No earlier result changes: entry 46's numbers are unchanged, and §36's caveat now states the measured reading. The registration counts in §21 and entry 43 rise by this study's three predictions and one checked result file.
- **Evidence.** Report §36; decision D35; `outputs/figures/hpsc_settling_real.png`, `outputs/hpsc_settling_REAL.json` (with the per-well table), `outputs/metrics.json` key `hpsc_settling`; tests `tests/test_hpsc_settling.py`.

51. Our features under another laboratory's definitions, and where they translate

Slide message: *Every spike-time record in the asset now also carries its features under the Kapucu 2022 depositors' stated definitions, from the same extractor; on their plates, three of seven features are the same number under both, three differ by a factor that depends on the culture (firing rate 1.1017 rat, 1.3902 human), and burst rate cannot be converted.*

- **What changed.** A registry of named definition sets, `src/neuromps/features/definitions.py`: `neuromps` (reference, the defaults the asset and the twin use), `kapucu2022` (validated, the settings ladder's S3) and `kapucu2022_logisi` (experimental, S4; used for no claim). Each is a bundle of `FeatureParams` options plus whether a deposit's noisy electrodes are dropped, so `features/mea.py` stays the one extractor. The ladder and the logISI study now take S3 and S4 from the registry; the definitions are equal to the old ones. A new stage in the models step (`src/neuromps/models/feature_definitions.py`, after the extractor agreement) re-reads each spike-time record with the ETL modules' own loaders (`src/neuromps/etl/spike_trains.py`), writes `data/processed/features_by_definition.parquet` (record id, source, definition, feature, value) and generates `outputs/tables/feature_definitions.md`. Decision D37; report §3, "Named definition sets"; `schema/STANDARD.md` §4.4.
- **Why.** D33 recorded this as the future option: a laboratory reading our numbers beside its own needs to know where a translation is supported by evidence, and the evidence (the ladder and the logISI test) was only in a metrics file.
- **Effect.**
- The sidecar holds 1,512 records under every set (40,824 rows). Its `neuromps` rows equal the asset in all 13,608 values; on the developmental plates the `kapucu2022` ratios against the depositors equal the ladder's S3 numbers in 28 of 28 comparisons (all checks: true).
- `kapucu2022` against ours, on 336 recordings: share of spikes in bursts, spikes per burst and burst duration are the same number (ratio 1.0132 for the first); active electrodes, firing rate and STTC are a scale factor (0.734, 1.3264 and 1.315); burst rate has no conversion (ratio 2.0584, rank agreement 0.9299).
- The factors differ by plate, so they are properties of these cultures, not constants of the definitions: active electrodes 0.9008 on the rat plate against 0.6958 on hPSC_MEA2.
- A same number between two of our sets is not agreement with the depositors: burst duration is unchanged by the switch, and `kapucu2022` sits at 0.8208 of their value.
- Not a registered study, so the registration counts (§21, entry 43) do not change. No record of `neuromps.parquet`, no method block, no twin target and no earlier key of `outputs/metrics.json` changes; the stage adds the key `feature_definitions` and costs a few seconds on the quick tier, which has no developmental-plate spike files and so skips the translation.
- **Evidence.** Report §3 ("Named definition sets"); decision D37; `schema/STANDARD.md` §4.4; `data/processed/features_by_definition.parquet`; `outputs/tables/feature_definitions.md`; `outputs/metrics.json` key

`feature_definitions`; tests `tests/test_feature_definitions.py`, `tests/test_definition_sidecar.py`, `tests/test_feature_translation.py`.

Work queue (decided 2026-10-24, from the post-round-1 review)

Item	Status	Registered before running
Synaptic depression under GABA-A block	done, entry 7 (failed as registered)	yes
ToxCast potency bridge	replaced by entry 6 (only four shared compounds)	—
Output-tract electrode	done, entry 9 (4 of 4 pass; 7 of 10 draws)	yes
Brain and nerve cord as a two-chamber chip (male CNS)	scoped out: one chamber pair is an anecdote, and FlyWire already tested 698 region pairs (§25)	—
Twin posteriors as classifier features	done as a twin-fitted prior, entry 8 (the delta posterior infers one knob, outside Task 1's domain)	yes
Random-graph-to-connectome blend for organoids	dropped	—
Embodied, closed-loop fly (body, nerve cord to muscles, sensory feedback)	roadmap after the final (`report/ROADMAP_EMBODIED_FLY.md`, entry 12); its first step, motor-neuron readouts, is entry 10	—

Why the blend was dropped: grafting fly wiring onto a model of a mammalian cortical culture has no biological basis, and organoid–monolayer differences have many confounds, so a match would prove nothing. The embodied fly is not dropped but scheduled: it adds new information only with the loop closed (nerve cord to muscles, body, sensory feedback), which is a research project for after the final; reading the motor neurons (entry 10) is its first step.